

PROJECT PRESENTATION

STUDENT INFORMATION SYSTEM

A computerized student record and certificate management system



FRONTEND

VB.NET

Desktop GUI developed with Visual Studio



BACKEND

MS Access

Relational database with OLE DB connectivity



CORE FUNCTION

Records & Docs

Centralized student profile and certificate tracking



Purpose

Computerize student record maintenance and reduce manual ledger-based work.



Core Records

Personal/student details, academic information and certificate/document paths.



Users

Administrative staff use the system through an admin login.



Quick Data Entry & Retrieval

Streamlined data input and instant retrieval of records to save time and administrative effort.



Flexible Data Viewing & Filtering

Student profiles can be easily searched, filtered, and organized by various key parameters.



ID-Based Document Access

Certificates and supporting documents can be viewed instantly using unique Student IDs.



Paperless Efficiency

Reduces reliance on physical ledgers, eliminating manual searching and paperwork backlog.

2. Problem Statement & Existing System



Existing Approach

Student information and certificates were maintained manually using notebooks and physical ledgers, including department-wise and academic-year-wise records.



Manual Ledgers

Handwritten records maintained in physical notebooks and paper files.



Departmental Organization

Records segmented separately across individual department offices.



Academic-Year Records

Archived batch-by-batch, making historical cross-referencing complex.



Key Drawbacks



Human Errors in Data Entry

High vulnerability to manual recording mistakes and typographical errors.



Time-Consuming Search

Manual ledger lookup and verification require significant time.



Slow Processing

Delays in administrative workflows and certificate generation.



Risk of Lost Records

Past paper records are vulnerable to loss, damage, or misplacement.



More Manpower & Paperwork

Heavy reliance on physical storage and continuous manual labor.



Unpresentable Reports

Handwritten reports lack professional structure and visual clarity.

3. Proposed System



Accuracy

Structured data entry and validation.



Speed

Faster search, retrieval and filtering.



Less Paper

Reduces dependence on physical ledgers.



Centralized Record Maintenance

Centralized computerized student record maintenance



Dedicated Module Forms

Separate forms for entry, viewing and certificate access



Multi-Criteria Information Filtering

Student information filtering by department, academic year, community and gender



ID-Based Document Viewing

Student ID-based certificate/document viewing



Input Validation Checks

Validation checks help reduce incorrect input



Quick Retrieval & Administration

Designed for quick retrieval and easier administration

4. Technology Stack & System Specification



Frontend

VB.NET Framework

Windows desktop GUI interface

Microsoft Visual Studio 2010



Backend

MS Access 2013

Relational database engine

OLE DB connectivity layer



Environment

Windows 8 OS

Development system specs:

2 GB RAM | 300 GB HDD storage



Application Forms & UI Components

Project archive includes dedicated VB.NET forms for:

- **Login & Dashboard** - User authentication & central navigation
- **Entry & Find** - Data input, updating, and multi-criteria filtering
- **Proofs / Certificates & Info** - ID-based document viewing & system details

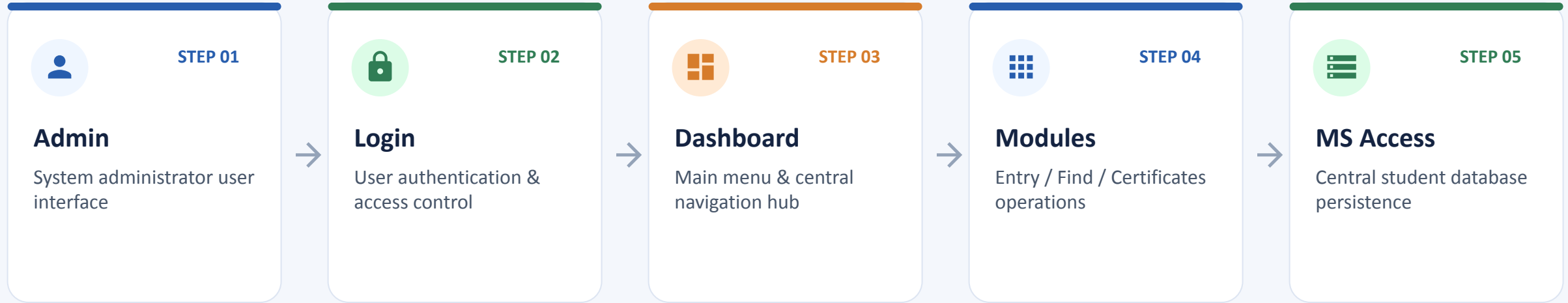


Database & Binary Artifacts

Complete project package includes:

- **Access Database File** (sis.accdb) - Stores student records & document references
- **Compiled Executable** - Production-ready desktop binary application

5. System Architecture / Flow



Key Architectural Flow Highlights



Sequential Execution

Login → Dashboard → Operational module



Entry Module Operations

Performs save, update, and delete actions



Find Module Capabilities

Supports multi-criteria filtering for student data



Certificate Viewer

Retrieves student documents using Student ID

6. Database Design



Student Table

Primary student record structure includes:

Personal Profile

Student ID, Name, Father's Name, Gender, DOB

Contact & Personal Info

Religion, Community, Address, Phone, Email

Academic Details

Department, Academic Year, Join Date, Roll No, Register No

Attachments

Document / Image Paths



Documents Stored

Digital file repository includes:

- Photograph
- 10th Marksheet
- 12th Marksheet
- 12th Transfer Certificate
- Aadhaar Image
- Passbook Image
- Community Certificate Image



Data Operations

System CRUD & search capabilities:

Core CRUD Actions

- Create / Save
- Update
- Delete

Direct Lookup

Search by Student ID

Multi-Criteria Filtering

Filter by Department, Academic Year, Community, and Gender

7. Main Modules



Admin Login

MODULE 01

Verifies the administrator before entering the system.

Authentication & Access



Student Entry

MODULE 02

Enter, save, update and delete student details.

CRUD Data Management



Student Information Viewer

MODULE 03

Filter and view records using available criteria.

Multi-Criteria Filtering



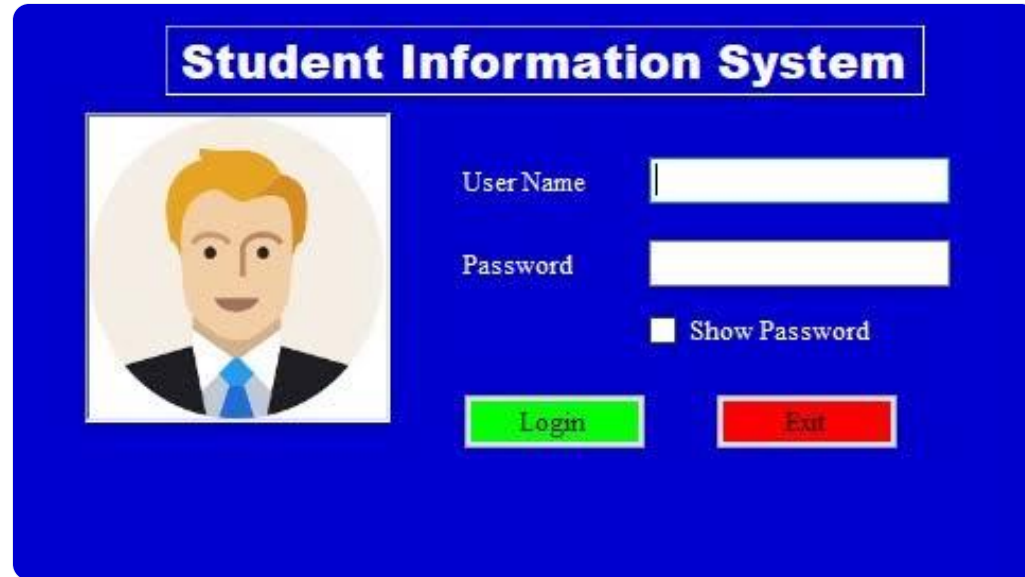
Certificate Viewer

MODULE 04

View student documents based on Student ID.

Document Retrieval

8. User Interface – Login & Dashboard



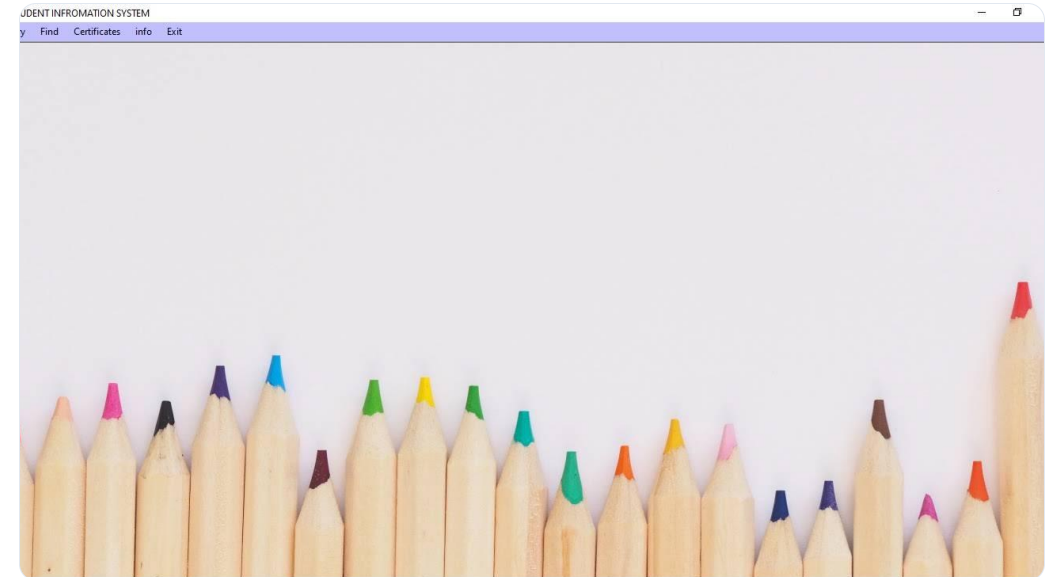
The image shows a login form for the 'Student Information System'. It has a blue background. At the top, the title 'Student Information System' is in a white box. Below the title is a circular placeholder for a user's profile picture. To the right of the picture are two input fields for 'User Name' and 'Password'. Below the password field is a checkbox labeled 'Show Password'. At the bottom are two buttons: a green 'Login' button and a red 'Exit' button.

MODULE 01 • AUTHENTICATION

Admin Login Form

Secure authentication portal for system administrators to verify credentials before granting access to student management modules.

Authentication & Access



OVERVIEW • SYSTEM HUB

Main Dashboard

Central navigation console providing quick entry points to Student Entry, Information Search, and Certificate Verification.

Navigation & Control Hub

9. User Interface – Data Entry & Viewer

STUDENT INFORMATION STUDENT ID: 5

PERSONAL INFORMATION

NAME:
FATHER'S NAME:
GENDER:
D.O.B: 08-10-2026
RELIGION:
COMMUNITY:
ADDRESS:
PHONE:
EMAIL:
BROWSE

ACADEMIC INFORMATION

DEPARTMENT:
ACADEMIC YEAR: FROM 08-10-2026 TO 08-10-2026
JOIN DATE: 08-10-2026
ROLL NUMBER:
REG. NUMBER:
BROWSE **BROWSE** **BROWSE**

OTHER PROOFS

AADHAR:
BROWSE

PASSBOOK:
BROWSE

COMMUNITY:
BROWSE

PROOFS FROM SCHOOL

10th MARKSHEET: 12th MARKSHEET: 12th TC:
BROWSE **BROWSE** **BROWSE**

OPERATIONS

CLEAR **SAVE** **UPDATE** **DELETE**

SEARCH

STUDENT ID: 2 **SEARCH**

MODULE 02 • DATA ENTRY

Student Entry Form

Comprehensive form interface for capturing student personal details, academic history, proof documents, and executing real-time CRUD operations.

CRUD Data Management

STUDENT INFORMATION

DEPARTMENT: ACADEMIC YEAR: COMMUNITY: GENDER:

SELECT OPTION

☐ DEPARTMENT
☐ ACADEMIC YEAR
☐ COMMUNITY
☐ GENDER
SEARCH

id	name	name	gender	dob	religion	community	address	phone	email	department	afrom	ayto	id	adatum	regnum
1	mayal	kannan	MALE	23-03-1999	HINDU	MB	south street, sakikanyakkannur	909898989	mayal@gmail.com	B.COM	01-06-2016	03-03-2019	01-06-2016	100	100
2	guru	murugan	MALE	11-05-2001	HINDU	MB	south street, sakikanyakkannur, nikkutta	7767876787	guru@gmail.com	BSc(CS)	01-06-2019	01-06-2022	01-06-2019	101	101
3	Niktha	thennarasu	FEMALE	08-03-2023	HINDU	MB	ayampalayam	6278978962	niktha@gmail.com	B.Com (CA)	22-06-2020	25-04-2023	26-06-2020	123	123
4	Abi	Murugan	FEMALE	17-01-2004	HINDU	MB	SKVR	1234567890	abi@gmail.com	B.COM	01-06-2016	03-03-2019	01-06-2016	1366	1266

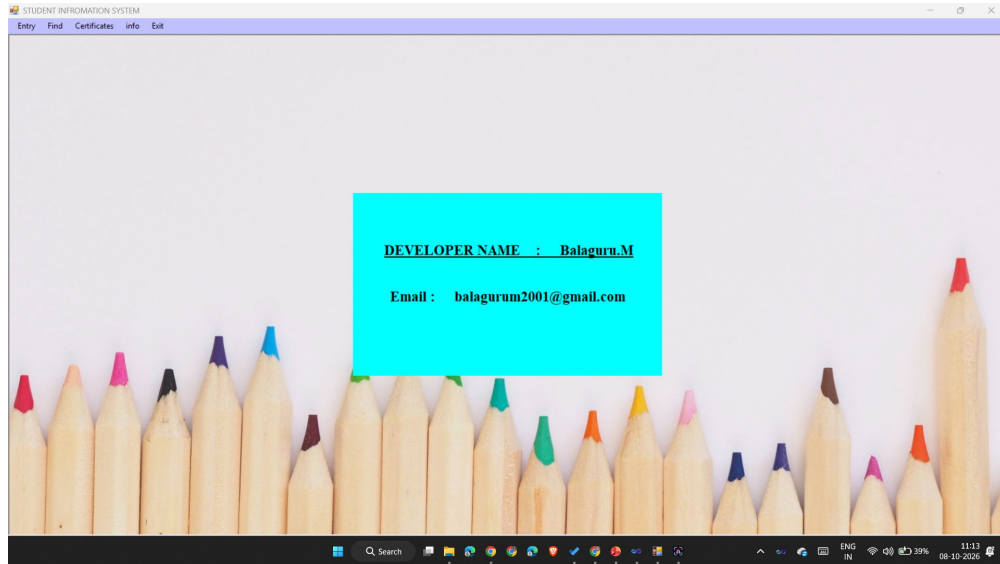
MODULE 03 • DATA VIEWER

Student Information Viewer

Interactive data grid with dynamic multi-criteria search and filtering by Department, Academic Year, Community, and Gender.

Multi-Criteria Filtering

10. Certificate Viewer & Testing



Certificate Viewer Interface

Interactive module for student certificate preview, verification, and testing validation.

METHODOLOGY

Testing Methods Documented

Includes Unit, Integration, Validation, White-box, and Black-box testing techniques.

MODULE COVERAGE

Test Cases & Scenarios

Comprehensive test cases documented for Login, Dashboard, Entry, Find, and Certificate modules.

VALIDATION

Reported Results & Verification

Reported test execution results demonstrate full alignment between expected and actual outcomes.

11. Implementation & Validation

01

Develop

Forms layout & design
Database connectivity setup
Core module logic implementation

02

Validate

Strict input data validation
Comprehensive error handling & feedback

03

Test

Module integration testing
Functional test cases verification

04

Deploy

Executable desktop application
Deployment for administrative use

12. Conclusion & Future Enhancement

PROJECT SUMMARY

Conclusion

Converts manual student-record administration into a fully computerized, automated workflow.

Key Impact & Outcomes



Enhanced Accessibility

- Centralized data allows instant retrieval of records anytime.



Reduced Clerical Workload

- Eliminates redundant paper handling and manual verification steps.



Simplified Management

- Supports seamless maintenance of student profiles & certificates.

NEXT STEPS

Future Scope

Identifies clear opportunities for continued system evolution as technology standards advance.

Planned Enhancements



Advanced Feature Extension

- Incorporate automated reporting, analytics, and self-service portals.



Cloud & Web Deployment

- Transition from current desktop setup to cloud-hosted web access.



Continuous Optimization

- Adapt modules to emerging security and database technology shifts.

THANK YOU

Student Information System